

The HTML5 Architect

Designing & Building Interactive Web Pages|

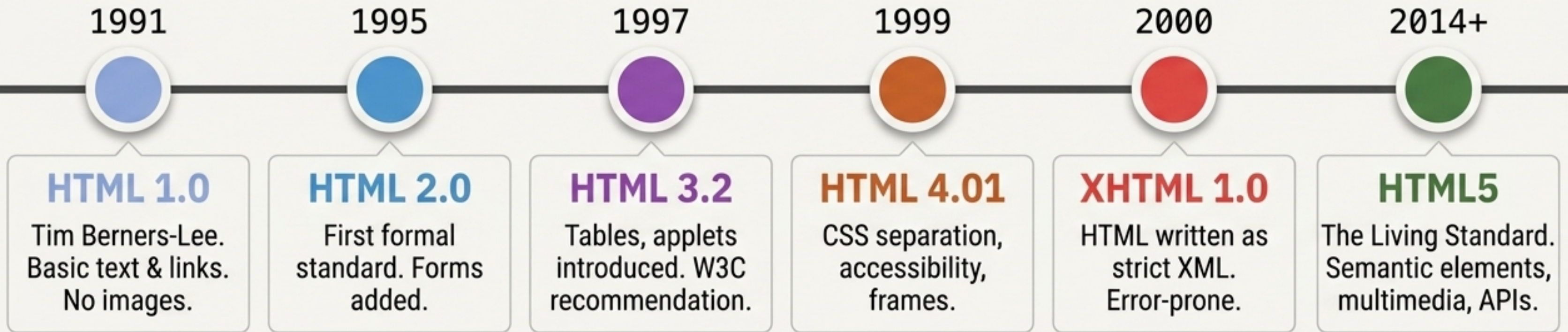


// Copperbelt University / CS 361 Module 2 / Dr Z. Lifelo

2.1 What is HTML? – History & Versions

HTML = HyperText Markup Language

The standard language for creating web pages. It describes the STRUCTURE and CONTENT of a document using a system of ELEMENTS represented by TAGS.



★ HTML5 (2014–present) is the **LIVING STANDARD** – continuously updated by WHATWG. No version numbers – just 'HTML'.

The HTML Boilerplate – An Exploded Box Model

```
1 <!DOCTYPE html>
2 <html lang='en'>
3   <head>
4     <meta charset='UTF-8'>
5     <meta name='viewport' content='width=device-width,
6 initial-scale=1.0'>
7     <title>Document</title>
8   </head>
9   <body>
10     <h1>Hello, World!</h1>
11   </body>
12 </html>
```

`<!DOCTYPE html>`

Not a tag. Tells the browser: 'This is HTML5.' Mandatory first line.

`<html lang='en'>`

Root element wrapping the page. 'lang' attribute is critical for screen readers.

`<head>`

Invisible to the user. Contains metadata, charset, CSS links.

viewport meta tag

Critical for mobile responsiveness to prevent desktop-layout zoom out.

`<body>`

Everything the user SEES. Visible content, text, media, forms.

2.3 The Shift to Semantic HTML – Structure with Meaning

Old (Non-Semantic) HTML

```
<div id='header'>
```

```
<div id='nav'>
```

```
<div id='content'>
```

```
<div class='post'>
```

```
<div id='footer'>
```

Browser has NO IDEA what these generic divs mean.



Semantic HTML5

```
<header>
```

```
<nav>
```

```
<main>
```

```
<article>
```

```
<footer>
```

Structure with meaning. Improves accessibility, SEO, and developer readability.

Common HTML Elements – Component Reference

Headings & Text

```
<h1> to <h6>, <p>,  
<strong>, <em>
```

[SEO CRITICAL]

One <h1> per page. Use / instead of outdated /<i>.

Lists

```
Unordered <ul>,  
Ordered <ol>,  
Description <dl>
```

[STRUCTURE]

Lists ALWAYS need child items inside. Never nest content directly in or .

Hyperlinks (<a>)

```
href='https...'  
target='_blank'  
rel='noopener'  
href='#section'  
href='mailto:...'
```

Supports external, internal, anchor, and email links.

Media Elements & Constraints

Images (& <figure>)

```
<img src='img.jpg'
      alt='description'
      width='400'
      height='300'
      loading='lazy'>
```

alt is mandatory. width/height prevent layout shifts. <figure> wraps image and <figcaption>.

Audio (<audio>)

```
<audio controls>
  <source src='audio.mp3'
          type='audio/mpeg'>
</audio>
```

Use multiple source tags to ensure cross-browser compatibility.

Video (<video>)

```
<video controls
        poster='thumb.jpg'>
  <source src='vid.mp4'
          type='video/mp4'>
</video>
```

 **Avoid autoplay** - blocked by most browsers unless muted, and highly annoying to users.

HTML Data Tables: Structure & Accessibility

`<caption>` & `<thead>`

Contains column headings,
stays visible on scroll.

`<tbody>`

Main data rows, required
for DOM manipulation.

`<tfoot>`

Totals/summaries, structurally
pinned to the bottom.

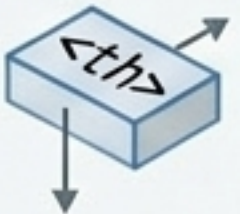
Header 1	Header 2	Header 3
Data A	Data B	Data C
Data A	Data B	Value X
Data A	Data B	Value Y
Total	Summary 1	Summary 2

`<tr>` (Row)



`<th>` (Header Cell)

`<th scope='col'>`



`<td>` (Data Cell)

`<td>`



For **TABULAR DATA** only. Never use tables for page layout.

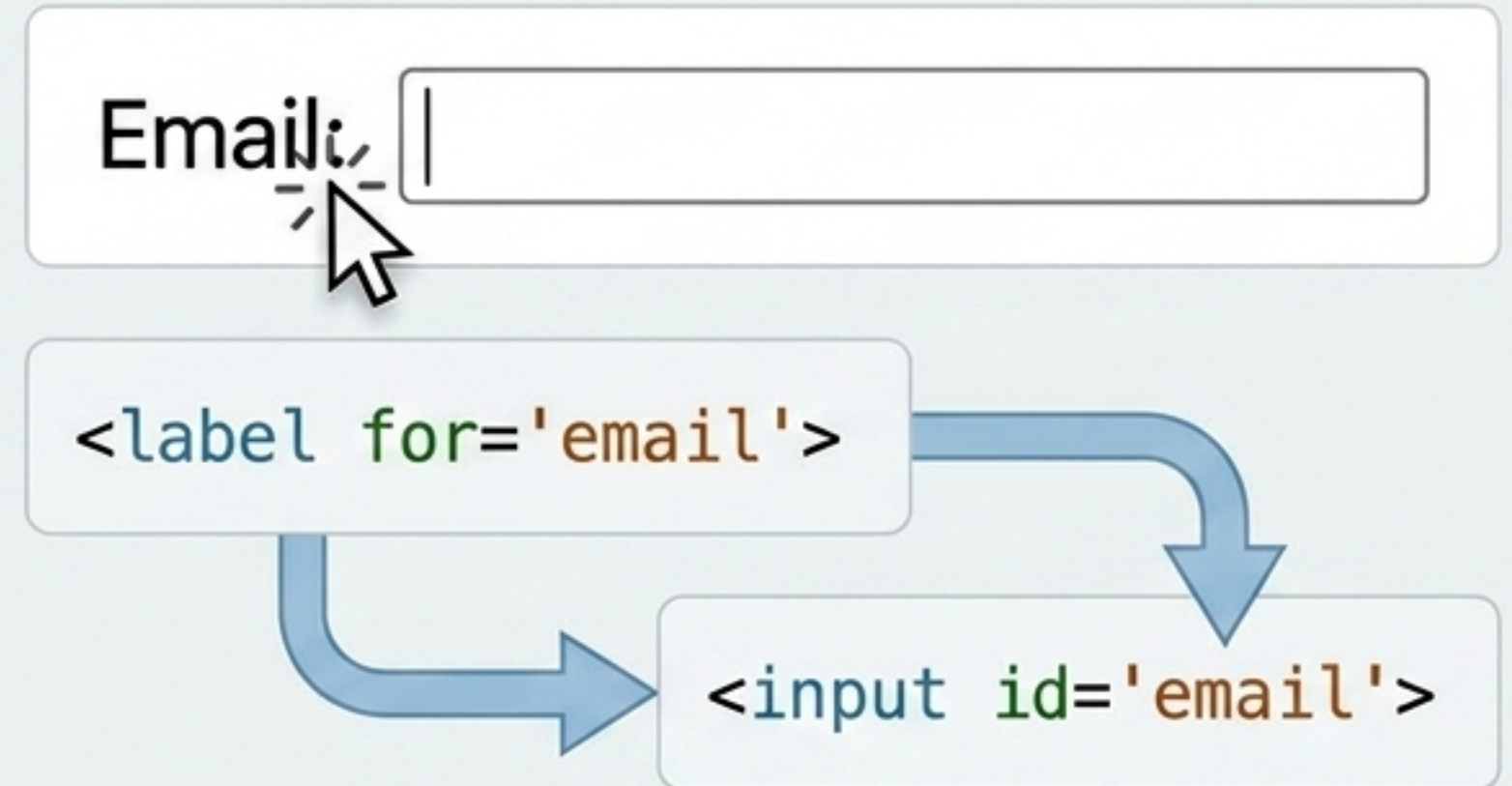
HTML5 Forms: Structure & Accessibility

The Form Architect

```
<form action='#' method='POST'
  enctype='multipart/form-data'>
  <fieldset>
    <legend>User Info</legend>
    <!-- inputs go here -->
  </fieldset>
</form>
```

Required for file uploads.

The Label Linkage Loop



1. **UX:** Bigger click target for users.
2. **Accessibility:** Screen readers announce the label on focus.

The HTML5 Input Catalog

Text & Formatting

`<input type='password'>`

`<input type='search'>`

`<input type='url'>`

`<input type='tel'>`

Metrics & Selection

`<input type='number'>`

`<input type='range'>`

`<input type='checkbox' checked>` ☒

`<input type='radio' checked>` ☒

Dates & Media

`<input type='date'>`

`<input type='time'>`

`<input type='color'>`

`<input type='file'>`

Validation Attributes

- `required`
- `pattern`
- `minlength`
- `maxlength`

No JavaScript needed!
Browser validates
on submit.

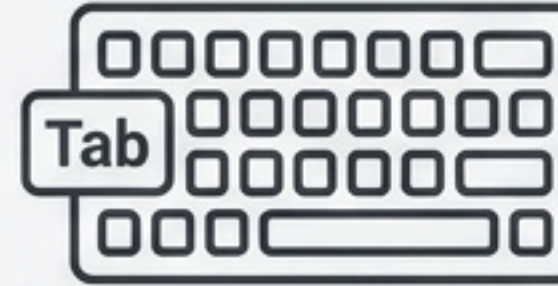
P - Perceivable



Code link:

- `'alt'` text on images,
- subtitles on video,
- **4.5:1** color contrast.

O - Operable



Code link:

- Functions usable by keyboard only,
- logical tab order,
- `'label'` click targets.

WCAG 2.1

U - Understandable



Code link:

- `<html lang='en'>`,
- clear semantic tags,
- predictable input validation.

R - Robust



Code link:

- Valid, error-free markup parsing correctly, across all modern browsers and assistive tech.

The Final Inspection

W3C[®] validator.w3.org - Zero errors = Valid HTML5.

✗ Missing <label> binding.

✓ Ensure 'for' attribute exactly matches input 'id'.
e.g., <label for='email'>Email:</label> <input id='email'>

✗ Missing alt on images.

✓ Add descriptive alt=''.
e.g.,

✗ Presentational tags.

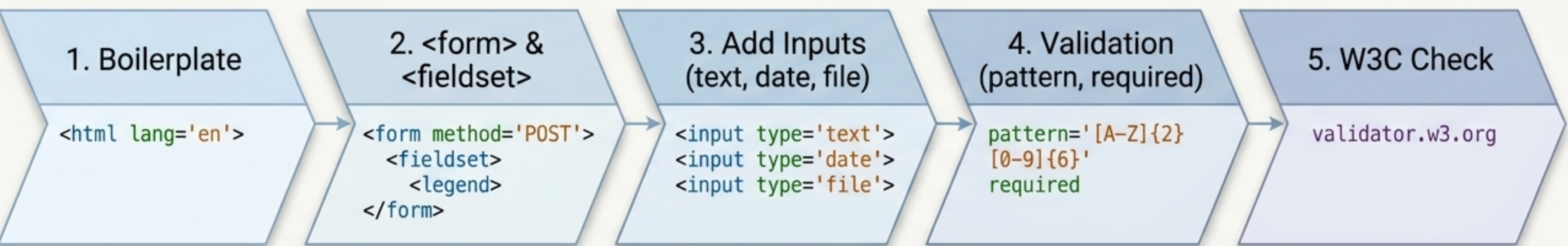
✓ Swap /<i> for semantic /.
e.g., Use Important, not Important.

✗ Malformed DOCTYPE.

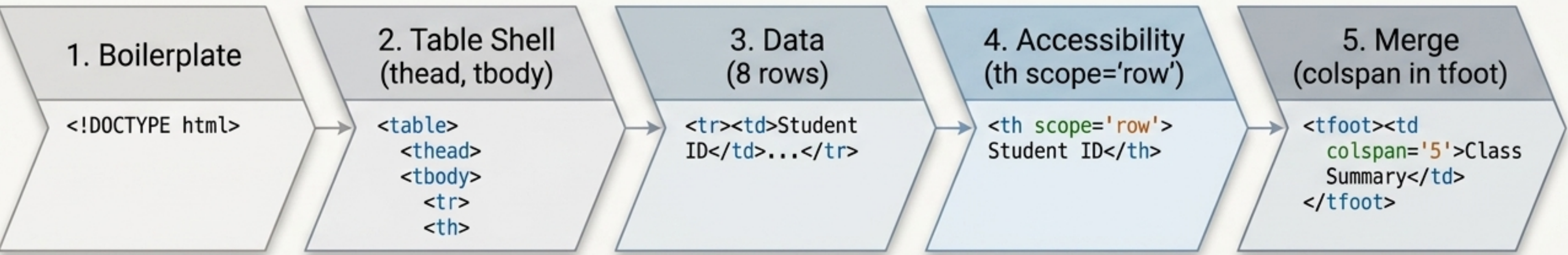
✓ Exactly <!DOCTYPE html> on line 1.
e.g., <!DOCTYPE html>

Week 4 Lab Activity: HTML Development Workflow

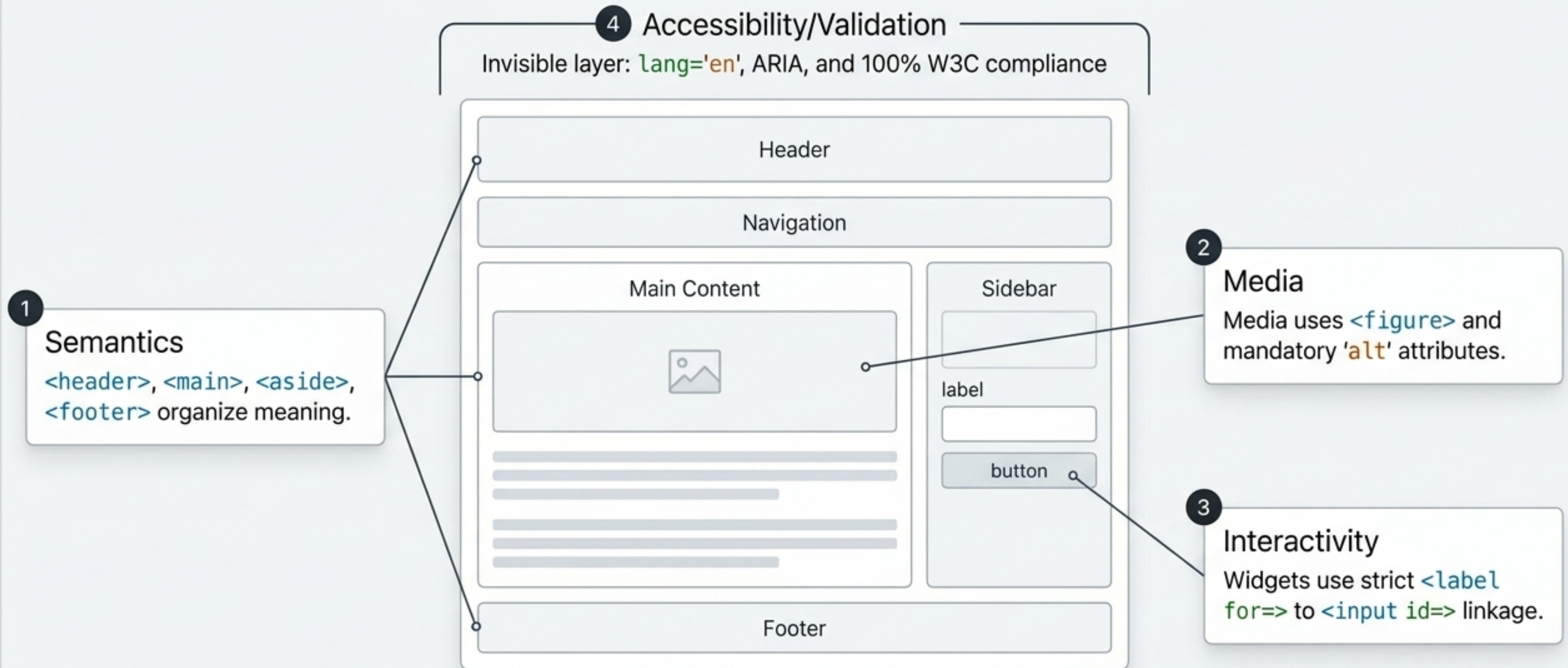
Track A: register.html



Track B: table.html



Anatomy of a Perfect Document



HTML is the skeleton. Mastering structure now prevents debugging pain later when adding CSS and JavaScript.